ENVIRONMENTAL SCIENCES
AT THE UNIVERSITY OF VIRGINIA

Across Grounds Collaborations

2023-24 Year in Review

Utqiaġvik, Alaska, is the site of a $3 million National Science Foundation project led by Professor Howie Epstein with contributions from researchers across the university.

The Department of Environmental Sciences

ESTABLISHED IN 1969, the University of Virginia's Department of Environmental Sciences was one of the first to look at fundamental environmental processes from a multidisciplinary perspective and the first in the nation to offer undergraduate, master's, and doctoral degrees in environmental sciences. Today, the faculty includes winners of the prestigious Tyler and Hutchinson awards as well as several professors who are among the most highly cited researchers in their fields.

Departmental field stations and facilities include the Anheuser-Busch Coastal Research Center in Oyster, Virginia, home of the National Science Foundation-sponsored Virginia Coast Reserve Long-Term Ecological Research program, the Virginia Forest Research Facility in nearby Fluvanna County, and the Blandy Experimental Farm near Front Royal, Virginia.

Pictured on the cover from left: Matthew Burtner, Professor of Music and Ajay Limaye, Assistant Professor of Environmental Sciences

PUBLISHER AND WRITER: Charlie Feigenoff DESIGN: Roseberries EDITING: Rosalind Hingeley PHOTOGRAPHY: Tom Cogill unless noted otherwise.

KEVIN SMITH / ALASKA STOCK

FROM THE CHAIR

The environmental sciences have historically focused on observation and analysis with the goal of understanding the natural world. We have now reached the point, globally and socially, where it is imperative that we think about how we can use that knowledge to solve the climate crisis and other environmental challenges that are rapidly altering life on our planet. The complexity of these issues—and the difficulty of developing effective responses—dictate that we take a cross-disciplinary approach. This year, our annual review highlights efforts faculty members are making to form partnerships across the University.

As you'll see, our faculty members have built bridges with colleagues in the humanities and social sciences, architecture and engineering, medicine and the data sciences. They have learned their languages and identified ways to collaborate on urgent problems. In many cases, they have also forged alliances with their counterparts at other universities, with government officials, and with citizen groups.

Faculty members' willingness to cross disciplinary borders has helped them secure large, high-profile grants from the National Science Foundation, including two from its Coastlines and People program and another from its Navigating the New Arctic initiative. The University's Environmental Institute, led by Karen McGlathery, has played an important role in promoting the formation of these groups.

Another way the department has jump-started this process is by hiring faculty with joint appointments at other schools at UVA or with cross-disciplinary expertise. They include Larry Band (with Engineering), Antonios Mamalakis (with Data Science), and Meghan Blumstein (with Architecture) as well as Charity Nyelele, who brings her expertise in community engagement to her environmental research.

In reaching across disciplines, we are modeling an approach to knowledge gathering and problem solving that will serve our students well. We hope that they will be inspired by our example—and the findings we bring to the classroom—to take a more nuanced and comprehensive view of the environmental challenges we all face.

Matthew A. Reidenbach

Matt Reidenbach
Chair

A Logical Progression

Although **Deborah Lawrence**'s career has taken her from the jungles of Indonesia to the halls of the State Department, her goal has always been consistent. "The common thread for my research has always been the rain forest, saving it and not just studying it," she says. "I became interested as a twenty-year-old and never wavered, but my view of what I could do broadened as time went on."

Applying a Climate Lens to Carbon

A biogeochemist, Lawrence was initially interested in how forests respond to disturbance and how the loss of nutrients affects their recovery, studying them on the ground in Mexico and Costa Rica as well as Indonesia. But early in her career, she began to view them from a larger, cross-disciplinary perspective. "In Mexico, I was interested not just in how the ecosystem responded but also in how its hydrology was altered and how atmospheric deposition changed," she says. "All these factors fed back into the way they grew, their ability to sequester carbon, and their capacity to provide for the humans who lived in them."

In 2009, Lawrence became a Jefferson Science Fellow at the State Department—and this experience gave her a new, even broader perspective. She served as science advisor in the climate office, concentrating on the impact of deforestation on climate. She was part of the U.S. delegation to the United Nations Framework Convention on Climate Change and to the World Bank's Forest Carbon Partnership. "Before I went to Washington, my focus was biodiversity and ecosystem function, forest conservation from the perspective of sustainable use," she says. "When I returned, I saw forests as drivers of planetary change, which completely changed everything I did."

Climate became the lens through which Lawrence viewed the natural world. She launched the first pan-University climate initiative—Food, Fuel, and Forests—which engaged more than a dozen faculty experts in law, engineering, commerce, anthropology, ethics, and policy as well as scientists from the National Center for Atmospheric Research.

Lawrence also became a more public scientist. She worked with governments and foundations around the world. She talked to journalists and spoke at TEDxCharlottesville. And she created a series of courses meant to engage undergraduates more deeply in climate issues. This included a popular introductory course on climate change, which, playing on the name of Al Gore's climate film, she called *An Inconvenient Truce: Climate, You, and CO2*. She launched a project called Write Climate, which encouraged students to express their thoughts and feelings about the changing climate and translate that into public art. And she revitalized the Environmental Thought and Practice major, a program that presents environmental issues within a framework that includes natural sciences, social sciences, and humanities.

"My point to students is that it doesn't matter what you study, you should be working on climate change," she says.

Taking the Next Step

As time went on, Lawrence became convinced that she could have more real impact on the course of the climate crisis and the fate of the rain forests outside the University. "I was sending climate warriors into the world, and I wanted to be one," she says. "I thought I could have a bigger impact by working on something that would directly connect my expertise and the climate crisis."

During her sabbatical in 2022, she began working with Calyx Global, a company whose goal is to restore integrity to the international carbon markets by rating the effectiveness of different carbon credit schemes. In 2023, she joined Calyx fulltime as its chief scientist.

"I loved my time at UVA and grew as a scientist because of the support I received from the department and my interactions with colleagues across the University," she says. "It set the stage for me to take this latest step." ■

Passing the Torch

As **Howie Epstein** notes, his appointment as the Sidman P. Poole Professor of Environmental Sciences at the beginning of the 2023–24 school year embodies a certain symmetry. The former holder of the chair, Jim Galloway, hired him. Epstein was department chair when Galloway retired and vacated the position. "The recognition means a lot to me," Epstein says. "To be considered in the same league as other endowed current and former professors in the department—Larry Band, Scott Doney, Karen McGlathery, Mike Pace, Hank Shugart, and Jim Galloway—is a real privilege." Epstein stepped down as department chair at the end of the year.

Opening the Arctic to Research

Epstein is a member of the generation of researchers who transformed and reenergized the study of the Arctic, an area that is warming four times faster than the global average. "When I started in the Arctic in 1996, I joined a relatively small community of scientists," he recalls. "Now, there are thousands of scientists trying to understand the implications of the massive changes sweeping the region."

An ecosystem and plant community ecologist, Epstein was among the first to trace the Arctic's greening. Between 2002 and 2012, he created a baseline by documenting vegetation along two north-south transects that extend far above the Arctic Circle, one in Siberia and the other in Alaska and Canada. He has been a lead author and regular contributor to the Tundra Greenness section of the *Arctic Report Card*, which is published each year by the National Oceanic and Atmospheric Administration.

Most recently, he received a $3 million grant from the NSF's Navigating the New Arctic initiative to launch a multidisciplinary effort to design and monitor a network of integrated meteorological, aquatic, and geotechnical sensors throughout Utqiaġvik, the northernmost settlement in Alaska. "The goal is not only to better understand the changing natural systems but also to assist the Arctic community in developing guidelines for buildings and infrastructure that increase their resilience while minimizing environmental impacts," Epstein says.

Recovering Landscapes

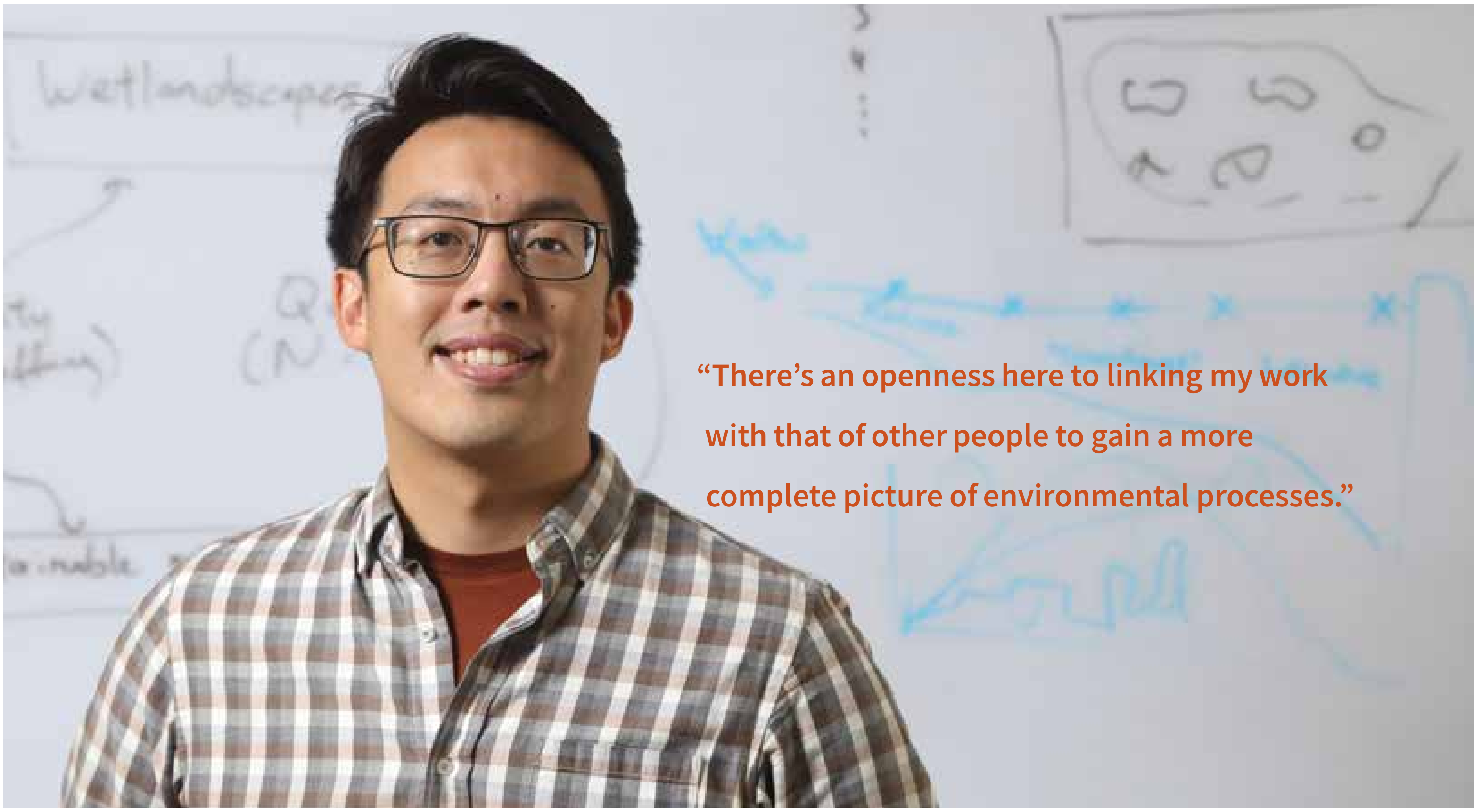
But Epstein has also maintained other research interests closer to home. He spent the last three decades studying successional dynamics and invasive species at the department's Blandy Experimental Farm. He is interested in such questions as the fate of farmland after it has been abandoned and the impact of invasive species on successional fields. "My appointment to a chair and associated research leave give me the opportunity to look back, synthesize my observations, and project how trends I've identified might unfold in the future," he says.

More recently, he has partnered with the Sisseton-Wahpeton Oyate community, part of the Santee Dakota people in South Dakota. Epstein and his students, primarily UVA undergraduates and local high schoolers, are studying the impact of grazing on the tall-grass prairie and assessing the efficacy of different restoration efforts. In addition, he is helping construct a community cultural center that will highlight student research while providing a place for community gatherings.

An Educator Above All

The educational component of these efforts is very important to Epstein. Although his research productivity has been impressive—he has authored almost 200 peer-reviewed articles—he says his biggest contribution has been the work he has done with students, both inside and outside the classroom. "Mentoring graduate and undergraduate students and widening their horizons is, in many ways, the most important thing that I am doing," he says. ■

"There are thousands of scientists trying to understand the implications of the massive changes sweeping the region."



Discovery at the Margin

Assistant Professor **Fred Cheng** focuses on areas of the landscape that are often ignored or misunderstood but are yet highly influential. These are the interfaces between the terrestrial and aquatic zones: the wetlands, the margins beside and underneath streams, and the ground between the surface and the water table. These areas are dynamic. Conditions vary depending on the amount of water flowing through them. They are also highly reactive. The materials and living organisms in these interfaces can change the water’s nutrient composition. “If, for instance, you want to gain an accurate picture of how nitrogen cycles through the environment and improves water quality, you need to have a clear understanding of what goes on in these interfaces,” Cheng says.

The Best of Both Models

Although Cheng works in concert with researchers in the field, his approach is primarily computational. This entails creating process-based models that emulate the physics affecting these interfaces as well as models derived from data-based machine learning techniques.

Each has its advantages and disadvantages. It can be difficult to fine-tune the parameters used in physical models, but because they are grounded in known processes, they can be extrapolated for use in similar circumstances. Machine learning can reveal underlying patterns, but because it is trained on localized datasets, it is difficult to apply its findings broadly. “Our goal is to get the best from each type of model,” he explains.

Counterintuitive Revelations

Cheng’s work has already revealed important insights. For instance, the estrogen fed to livestock to promote growth ultimately can make its way into streams. Prolonged exposure even to trace amounts can cause developmental issues or sex change in fish. While lab studies found that estrogens decay quickly in water, field studies revealed elevated and persistent levels of estrogen long after their application had ceased.

The answer to this conundrum, Cheng discovered, lay in the hyporheic zone of soil next to and below the streams. His physical modeling suggests that estrogen decay rates are suppressed in this sediment—and that relatively harmless estrogen by-products are converted back to more potent forms of estrogen. In other words, processes in the hyporheic zone counteract the decay of estrogen in streams.

A critical challenge for researchers studying these reactive interfaces is moving from small-scale analyses to large-scale generalizations, discovering patterns that seem to be true across a wide range of watersheds and conditions. Cheng has made progress here as well. One reason wetlands have received federal protection is that they absorb excess nutrients from the water passing through them, but revisions to the law have limited this protection to wetlands directly connected to large bodies of water.

Analyzing the data and metadata from many different studies of individual wetlands, Cheng found that small, isolated wetlands were actually more biogeochemically reactive than a single wetland of comparable size and could collectively remove more nutrients. “Based on our findings, we believe that spatially targeted wetland restoration can yield significant reduction in nitrogen transport,” he says.

Cheng joined the department in January 2024 and is excited about pursuing his research in an institution that values interdisciplinary collaboration. “There’s an openness here to linking my work with that of other people to gain a more complete picture of environmental processes,” he says. ■

Helping Forests Help Us

Assistant Professor **Meghan Blumstein** is worried about forests—and rightly so. "Forests are a huge part of the global carbon cycle," she says. "They take up carbon dioxide through photosynthesis, but they also respire, which means they release carbon dioxide." On net, forests take up more carbon dioxide than they release, she explains, making them a key component of global carbon sequestration. But the stresses of climate change—extreme weather, warming temperatures, and invasive pests and diseases—may turn this relationship on its head. "It's a huge question," she says. "My goal is to find ways to help forests better withstand these stresses and preserve their role as a carbon sink."

At the same time, Blumstein is interested in helping people who work in plant nurseries choose plants that are appropriate for the changing climate. As a result, she has a joint appointment in the Department of Landscape Architecture. "I want to make it possible for people who actually go out and plant trees to make better decisions when they design a landscape," she says.

Nature and Nurture

Critical to this effort is determining what combination of nature and nurture affects a plant's ability to withstand stress. "To the extent that nature plays a dominant role, our strategy will be to change seed stock or breed new crosses," she says. "If it's nurture, our goal will be to apply different management techniques." This might, for instance, entail inducing drought stress when a plant is young to bolster its tolerance.

In studying the natural component of adaptation, Blumstein has an advantage. With a doctorate in organismal and evolutionary biology, she has a range of genetic and genomic tools at her disposal. This includes genome sequencing, which can help her isolate meaningful genetic differences between those members of a tree species that are adapting to climate change and those that are not.

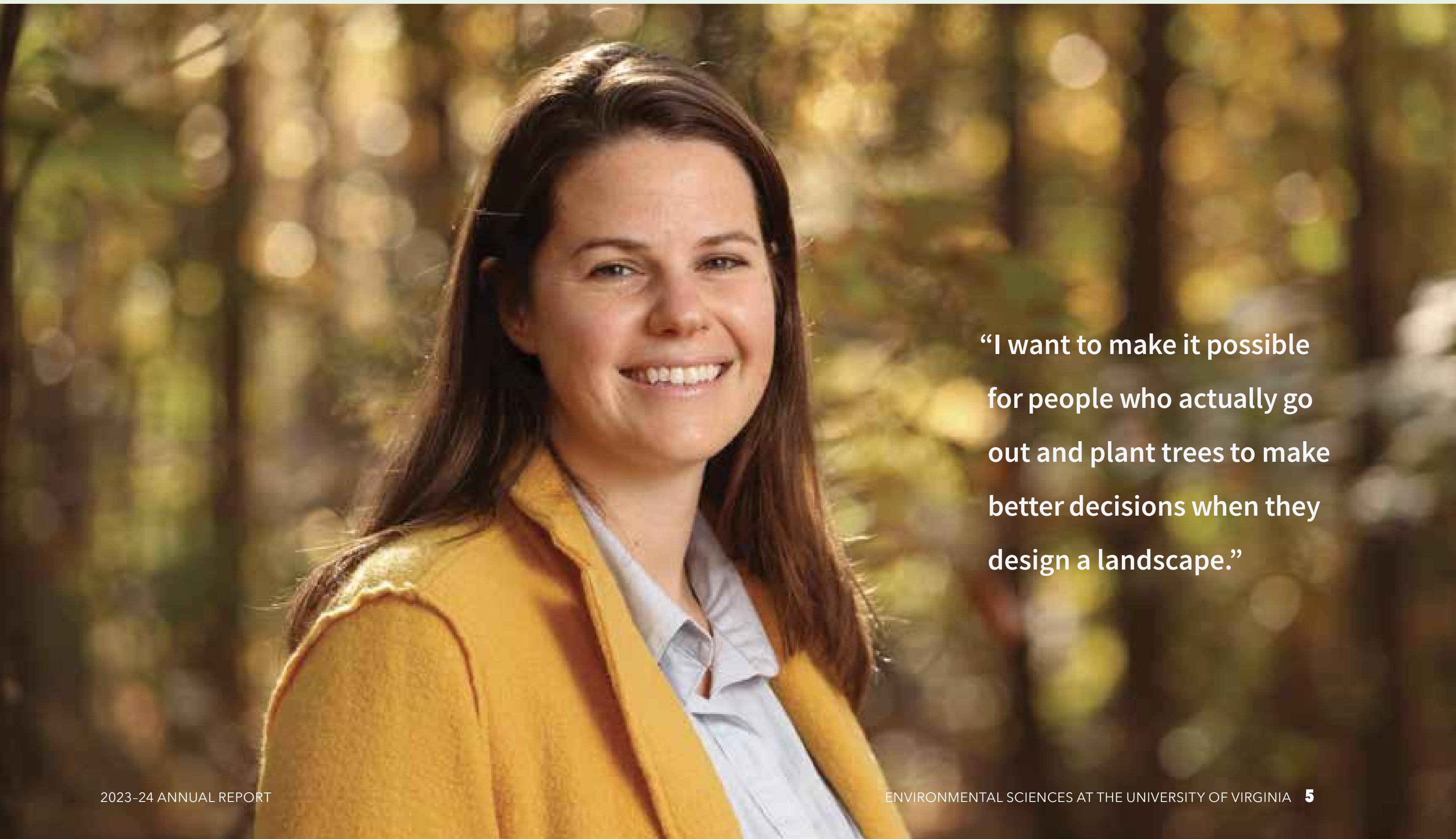
Looking at Red Oak

Blumstein is taking advantage of the University's decision to dedicate Morven, a nearly 3,000-acre estate in Albemarle County, as a sustainability lab. She is planting several acres of red oaks using acorns gathered from populations along the East Coast, from Alabama to Canada, to identify those that are best adapted to Piedmont Virginia and determine if genetics are driving their adaptability. "I've chosen the red oak to study because it's the most widespread oak across the eastern United States and a main driver of carbon sequestration," she says. "It is starting to decline, and no one knows why."

Colleagues are developing comparable sites in Maine, Massachusetts, and Alabama. "We hope to learn how best to reforest agricultural sites that are no longer being used," she says.

At the same time, Blumstein is collecting leaf tissue from the red oak's entire range and sequencing each sample's genome and determining how environmental conditions affect its growth.

Although she was hired in 2023, Blumstein arrived in Charlottesville at the beginning of the 2024–25 academic year and lives at Morven. "Everyone is super-supportive and sees the value of having more interactions between the department and the School of Architecture," she says. ■



"I want to make it possible for people who actually go out and plant trees to make better decisions when they design a landscape."

Launching Research at the University's Sustainability Lab

> "We are creating an access point for the broader University community to identify and investigate research questions across a variety of fields."

When media mogul John W. Kluge gave the University his extraordinary collection of farms and estates in southeastern Albemarle County in 2001, it more than doubled the University's land holdings. Since that time, the University has disposed of part of these holdings to create endowments to support estate maintenance and programs at Morven Farm, Kluge's 3,000-acre home. In 2022, it reenvisioned Morven as UVA's Sustainability Lab in support of the University's strategic commitment to environmental resilience and sustainability. Professor **Manuel Lerdau**, who has long been involved in Morven's programs, was named its director of research.

"Although environmental sustainability will be central to the research and educational initiatives we pursue at Morven, we are interpreting our sustainability mandate broadly," Lerdau says. "For instance, we're thinking about wellness as a component of individual sustainability." Lerdau is working with **Dr. Christopher Holstege**, the associate vice president of student health and wellness, and his colleagues in the School of Medicine to develop ways for Morven to serve as a center for such activities as contemplation and contemplative studies.

Lerdau is also thinking about Morven in terms of what he calls "cultural sustainability," honoring the ancestral land practices of the Indigenous nations who lived, hunted, and foraged on the land that Morven now occupies. Morven has built relationships with the chief and the resource manager of the Monacan Indian Nation—and plans to gradually return its loblolly farm plantation to the mixture of hardwood forest and open pine savannah that the Monacan maintained through burning. There is also interest in reestablishing the pawpaw tree, which the Monacan prized for its fruit.

Setting the Stage

Lerdau points out that for an organismal ecologist like himself, Morven is an incredible resource for studying such topics as photosynthesis and metabolism, but he plans to encourage an ever-greater range of colleagues to take advantage of the research opportunities that Morven offers. In summer 2024, he collaborated with Assistant Professor **Justin Richardson** to establish 90 long-term plots for soil and vegetation monitoring. "One of the real unknowns is how climate change affects ecosystems over long periods of time, not just the one to three years of the average grant, but over 10 years or more," Lerdau says. "We see Morven as a place to do that kind of decades-long research."

Lerdau is helping to provide a basis for future research at Morven by working with the Scholars' Lab to create a Morven remote sensing database that would be available to University researchers as well as the public. "This initiative takes advantage of the expertise of people in our department such as Professors John Porter and Xi Yang," Lerdau says. "By mapping Morven, we are creating an access point for the broader University community to identify and investigate research questions across a variety of fields."

For Lerdau, getting more people from across the University and beyond to see in Morven opportunities to pursue their research has been one of the most satisfying aspects of his position. "I find the potential of a diverse and dynamic research community based at Morven to be very exciting," he says. ■

Insight into an Invisible World

Despite his years of experience and his expertise, Professor **Steve Macko** has retained his sense of wonder about what can be revealed by specialized mass spectrometers and other analytic devices. "You can come into my lab and answer questions that can't be addressed in any other way," he says. "You can learn things about materials that otherwise might be reluctant to give up their secrets."

His equipment has given Macko the ability to makes discoveries in environmental science—for instance, he was able to confirm doctoral student Bob Swap's hypothesis that dust essential for the fertility of the Amazon has its origins in the Sahara Desert. Macko and another doctoral student, Tom O'Donnell, scanned the shell of a mollusk living 20,000 years ago and drew conclusions about the Earth's changing environment.

"Using extremely small samples, I can shed light on questions that arise in each of the environmental sciences," he says. But his equipment has also given him the tools to range far afield. Having examined a sample of Edgar Allan Poe's hair, he learned that fish was a major part of the writer's diet.

But feeding samples into a spectrometer is, in many respects, the least of what Macko does. For many of the challenges he takes on, Macko has to master their context and then determine how spectrometers can be used to shed light on the problem. For instance, he developed a tool that uses mass spectrometry to prove that certain molecules found in meteorites have their origins in space. "The community challenged us, and we developed a tool that people recognized as the only way to answer that question," he says.

Creating an Unmatched Assemblage

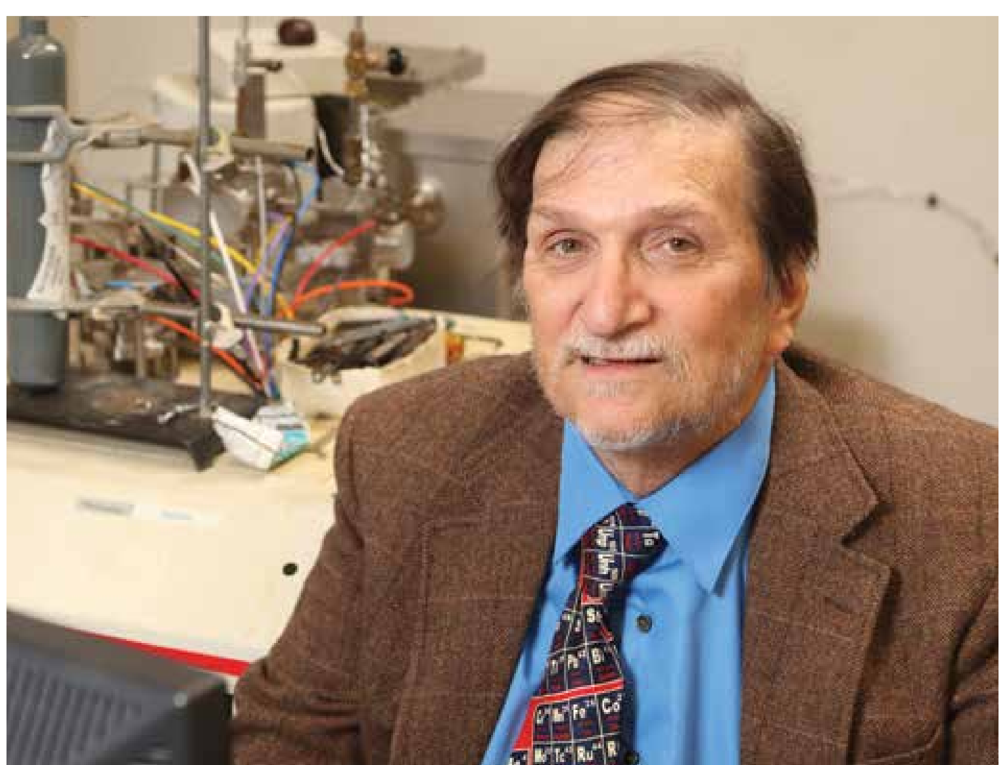
Macko has not only gathered a range of spectrometers for his own research, but he has played a leading role in securing funding for cutting-edge equipment—costing a million dollars or more—that faculty at the University and at other colleges in Virginia can use. He collaborated with University chemists and materials scientists to win a grant from the National Science Foundation for an X-ray photoelectron spectrometer (XPS), which is used to understand changes in surfaces undergoing chemical reactions.

The XPS can be deployed for tasks as varied as helping chemists tailor catalytic surfaces to promote the synthesis of specific chemicals as well as to study the chemical formation of organic molecules and investigate the origins of life. Macko's graduate student Christina Buscher used the XPS to examine the effects of ocean acidification on the carapaces of Atlantic blue crab larvae. She found that changes in the carapaces's chemical composition affected the crabs' likelihood of survival.

One reason the NSF responded favorably to the grant proposal is that the Engineering School has a centralized Nanoscale Materials and Characterization Facility with a dedicated staff that maintains the equipment. Macko has nothing but praise for **Cathy Dukes**, the principal scientist responsible for the XPS and other instruments.

Macko and his colleagues were recently selected by the University to submit a grant application to NSF for a time-of-flight mass spectrometer. This device will enable researchers to characterize the composition of extremely small samples more accurately. "We are always looking at the latest equipment because there are always new questions to ask," Macko says. "Having a time-of-flight mass spectrometer would make us a great candidate to examine extraterrestrial material, enabling us to shed light on conditions on Earth before the development of life." ■

"You can come into my lab and answer questions that can't be addressed in any other way."



AMELIE BERGER

"It's imperative that we train students to think outside of their disciplines."

A Machine for Collaboration

The challenges currently facing humanity are so broad and complex that their solutions require collaboration across a host of disciplines. Finding solutions is so urgent, however, that those universities and other organizations tasked with finding them are no longer waiting until these interdisciplinary partnerships arise organically. Instead, they are encouraging their formation deliberately and systematically.

That is the fundamental premise behind UVA's Environmental Institute (EI). "The institute was founded to make connections across Grounds that would otherwise not happen," says **Karen McGlathery**, the Sherrell J. Aston Professor of Environmental Sciences and the institute's director. "By bringing people together, you create ideas and pathways that catalyze new, essential, actionable research."

A Calibrated Set of Initiatives

Originally founded in 2017 as the Environmental Resilience Institute, the EI expanded its brief when the University made environmental resilience and

Identifying Groups Uniquely Vulnerable to Flooding

In 2022, the University of Virginia and Norfolk State University were awarded a $5 million grant as part of the National Science Foundation's Coastlines and People (CoPe) initiative to help researchers and Norfolk area residents respond to climate change. This is an interdisciplinary effort, involving experts from architecture, civil and environmental engineering, African American and African studies, social work, and environmental sciences—and graduate student **Heather Christensen** is in the thick of it.

"I have always been interested in natural hazards and risks to humans, particularly those people who might experience risk differently," says Christensen.

For her dissertation, Christensen is mapping the hydrological potential for flooding in various parts of Norfolk and the flood risks for the general population, as well as those who, like Christensen, have a mobility impairment. "Studies of hurricane response show that people with disabilities are less likely to evacuate before a storm hits and more likely to be stranded afterwards," she says. Such work is particularly important in Norfolk, where flooding from extreme storms is exacerbated by the region's rapid sea level rise, which often results in sunny-day flooding during high tides.

Christensen is advised by Professor **Matt Reidenbach**, one of the investigators on the project and an expert in hydrodynamics. In addition, she confers frequently with Professor **Jon Goodall**, a member of the Department of Civil and Environmental Engineering and the project's principal investigator.

Building a Map, Layer upon Layer

As she constructs her risk impact map, Christensen taps her mentors' complementary expertise. From Reidenbach she gains insight into how sea level rise, combined with such factors as soil, slope, and land cover, translates into risk. From Goodall and his team she learns how the built environment might mitigate or exacerbate that risk.

Christensen also consults census data that might help her locate mobility-impaired groups. In some cases, people having sustainability one of the five research themes of its 2030 Strategic Plan. Over time, the institute has developed a mutually reinforcing array of initiatives designed to nurture collaboration. Among other efforts, it convenes workshops, meetings, and forums. “We reach out to people around Grounds who are tackling a specific issue from different perspectives and invite them to have a conversation,” McGlathery says. “Through conversations and brainstorming, these teams begin to coalesce around a project.”

The institute also has established a variety of funding streams that it uses to promote the formation of these research groups. They include rapid grants for time-sensitive needs, Spark Grants for researchers work-shopping promising ideas with colleagues, and CoLab grants to help teams secure external funding.

This strategy has already yielded results. In 2020, the National Science Foundation (NSF), through its Navigating the New Arctic initiative, awarded $3 million to an interdisciplinary team led by environmental sciences professor **Howie Epstein** to design and monitor a network of integrated meteorological, aquatic, and geotechnical sensors throughout Utqiaġvik, Alaska. This effort started as a CoLab. The institute also brought together interdisciplinary teams that secured two $5 million NSF Coastlines and People awards to improve coastal resilience in Virginia.

As part of its environmental resilience and sustainability theme, the University is making even greater investments through EI to provide the type of funding critical to making transformative change in communities confronting climate issues. It has allocated $10 million for the EI’s Climate Collaborative program, which supports large, interdisciplinary research teams that work with policymakers, business leaders, government officials, and community groups to help localities take on climate-related challenges. Climate Collaboratives are exploring how Appalachia could benefit from the clean energy revolution and how women in India can be supported as change agents for water security in areas with increasing flooding.

Changing Academic Culture

In effect, the Environmental Institute has built a machine to promote the interdisciplinary research needed to focus and intensify the response to climate change. To date, it has provided funding for over 90 projects, ranging from building resilient urban water systems to assessing the effectiveness of hurricane evacuation orders.

But it has also played an important role in normalizing interdisciplinary research itself. In the 20th century, research was conducted within the framework of individual disciplines, and researchers were rewarded accordingly. It was risky for young faculty members to venture beyond their fields. “Although the culture at the University has changed dramatically in recent years,” McGlathery says, “the Environmental Institute, by promoting cross-disciplinary opportunities and celebrating successes, is helping to accelerate this paradigm shift.”

The Environmental Institute has also launched programs that ensure that the next generation of environmental leaders views interdisciplinary work as the norm. These initiatives span everything from undergraduate internships and externships to graduate and postdoctoral fellowships. “It’s imperative that we train students to think outside of their disciplines as well as outside of the box,” McGlathery says. ■

disabilities might disclose their condition, but she also makes use of indirect inferences. For instance, census tracts with a high percentage of people 65 and older are likely to have more people with disabilities than areas with a younger population.

She also consults the literature to find out how previous researchers have addressed similar challenges. One factor that jumped out at her from her reading was sidewalks. “When we visited Norfolk, we observed that many neighborhoods lacked sidewalks—and we saw from case studies in the United Kingdom that sidewalks helped people escape floods.”

As Christensen adds more layers of information to her map, she uses a method called geographic information system (GIS) multicriteria decision-making. She assigns values to the information in each layer depending on how it contributes to vulnerability. For instance, she might give a low number to areas with sidewalks and a high number to areas without them. She then combines the values of each overlapping

“Sidewalks helped people escape floods.”

layer to determine overall vulnerability for people with mobility impairments at any given spot in Norfolk.

“My goal is to create a map that people in the community can use to inform decisions about where they would prefer to live and that emergency responders can use to make choices about allocating life-saving resources,” she says. “And I would like to make the map as transparent as possible, so that users can change the weightings to better reflect their local knowledge.”

Gathering all this information and extrapolating from individual neighborhoods is a time-consuming process. This summer, Christensen secured an Environmental Futures Fellowship from UVA’s Environmental Institute, which enabled her to recruit an undergraduate, **Rachel Weghorst**, to help her with this work. ■

"Regardless of the landscape, channels seem to converge on certain patterns."

Mastering the Notation of Rivers

From a research perspective, the human affinity for seeing patterns can be a weakness as well as a strength. This is certainly true for the study of rivers. The basic building block of a meandering river is its bends, which are categorized by shape. Some seem symmetrical like bell curves; others are skewed like bananas. "As a field, we have adopted the general idea that there seem to be certain shapes in these bends that occur again and again and that drive rivers to shift across landscapes," says Assistant Professor **Ajay Limaye**. "The qualities associated with these shapes are used for such engineering applications as the siting of bridges and river restoration projects."

But, in fact, there has been no consistent way to determine if these particular shapes actually exist—or even where these bends begin and end. Limaye received a prestigious CAREER Award from the National Science Foundation to develop computing methods to decode the actual shapes that make up meandering rivers. "Our objective is to let the data do the talking and let rivers tell us through their own maps if these shapes are real or manifestations of our own imagination," Limaye says. "We are trying to distill rules that geologists have implicitly used going back decades to develop a new taxonomy driven by the data."

One advantage of being able to define bends precisely is that it will enable Limaye and his colleagues to identify gaps in models of how rivers change and evolve under different conditions so that they better match observations from the field. Another reason that Limaye is excited about this research is its wide application. "Meandering channels show up in different locations on Earth, whether in bedrock river canyons or lava flowing from volcanos," he says. "But they also appear in other places, for instance, on Titan, a moon of Jupiter, where liquid resembling gasoline flows over ice. Regardless of the landscape, channels seem to converge on certain patterns." Limaye believes his research can help explain this convergence while highlighting dissimilarities caused by different materials and geologic settings.

The Soundscape of a River

While Limaye's research is motivated by the need to understand patterns in the landscape, his new approach to the data also allows him to explore rivers' expressive qualities through sound. He is collaborating with UVA music professor **Matthew Burtner**, an Emmy Award-winning composer, sound artist, and ecoacoustician. Burtner has built a body of work based on transposing inputs from the environment into musical compositions. This approach immediately resonated with Limaye, a multi-instrumentalist himself.

Over the last year, Burtner and Limaye have been transforming the data by using sound to communicate the shapes of rivers. "We have a first composition, and we hope by the end of the year to have a full set of rivers transformed into music," Limaye says. "Our goal is to present them at a concert at Old Cabell Hall."

Limaye and Burtner are also working with UVA associate professor of education **Jennifer Chiu** to understand how audiences respond to the experience and help people engage with the science of rivers in a new way. ■

Tracking the Impact of Weather on Health

"If the data quality is what we think it is, the number of questions we can answer is tremendous."

Professor **Bob Davis** is the first to acknowledge that he cannot pursue the intersection of weather and climate with human health—his long-term research interest—on his own. "I bring to the table a pretty strong knowledge of climatology and a solid understanding of how stressors affect the human body," he says, "but I'm neither a physician nor an expert on healthcare policy." His limits are particularly evident, he says, when it comes to interpreting healthcare datasets. "Morbidity datasets include records of things like emergency department visits, hospitalizations, and doctor's office visits," he says. "It takes an expert to understand their nuances."

Tapping the Power of Cross-Disciplinary Expertise

A number of his cross-disciplinary research projects have received support from the Environmental Institute. Collaborating with **Dr. Kyle Enfield**, an associate professor in the Division of Pulmonary and Critical Care Medicine at UVA, Davis sought to determine if there is a correlation among weather, climate, and emergency room visits. Using data from 2010 to 2017 from UVA Health and the Carilion Clinic, Davis and his colleagues found that the risk of emergency room visits in Charlottesville and Roanoke increased 5 percent on warm humid days. They found that cold conditions kept people at home, but that visits increased 4 percent two to five days after a cold snap.

More recently, **Wendy Novicoff**, a professor of orthopaedic surgery and public health sciences at UVA, has secured access to a dataset that includes every interaction that Virginia residents have had with the state healthcare system over the past eight years. It contains 600 million records. Novicoff and Davis teamed up with **Pam DeGuzman**, an associate professor in the School of Nursing and an expert on healthcare disparities. Together they received Environmental Institute funding to recruit a postdoctoral researcher, **Truc-Ly Le-Huynh**, to help them analyze this information and correlate it with data from the Centers for Disease Control Social Vulnerability Index and the U.S. Census.

"When you are starting a new project like this, just getting the data organized to the point where you can do something useful can take months," Davis says. "Having someone to focus on this is critical." They are also enlisting the help of Assistant Professor **Antonios Mamalakis**, who has a joint appointment in data science and environmental sciences, to work on the project.

"If the data quality is what we think it is, the number of questions we can answer is tremendous," Davis says. "For instance, how much does a heatwave influence doctors' visits or prescription drug purchases? What are the impacts on people who have some level of disease that's not acute?" With this dataset, Davis feels that it will be possible to go beyond measuring extremes like morbidity and assess the net total impact of weather and climate on health.

In all these projects, Davis' goal is not simply to find relationships between weather and mortality and morbidity but also to drill down to root causes and identify particularly vulnerable populations. "Understanding root causes is essential to helping the medical community develop ways to mitigate the health impacts of climate change and extreme weather on all groups," Davis says. ■

Awards, Appointments, & Publications

Undergraduate Students

The The department recognizes fourth-year students who have done outstanding work in specific environmental sciences. This year, the Michael Garstang Atmospheric Sciences Award went to **Isabella Dressel** and the Mahlon G. Kelly Prize in ecology to **Olivia Beidler**. The department presented its Hydrology Award to **Kendall Colenbaugh** and the Wilbur A. Nelson Award in geosciences to **Kate McCarthy** and **Adelle Thrush**.

The departmental Interdisciplinary Award for the undergraduate major who has excelled in interdisciplinary environmental sciences research was presented to **Rachel Lombardo**. She also was this year's recipient of the Trout Unlimited Award. Established by the Thomas Jefferson Chapter of Trout Unlimited, this award is presented for "significant contributions to research concerning cold-water fisheries or related ecosystems."

Kendle Schooler was selected to receive the Hart Family Award for Undergraduate Research in Environmental Sciences. It provides funds to assist full-time environmental sciences majors who are conducting a supervised research project.

Victoria Thompson received the Wallace-Poole Prize, awarded each year to the graduating student majoring in environmental sciences who has at least a 3.8 GPA and who is judged the most outstanding student in the class.

The Bloomer Scholarship, which provides $1,800 toward tuition, is given to an outstanding undergraduate environmental sciences major with a focus on geology. This year's winner was **Marion Donald**.

Maya Weiss received the Richard Scott Mitchell Scholarship, which provides $1,800 to a rising fourth-year student who is focusing on geoscience and has completed Fundamentals of Geology and two other advanced courses in geoscience, preferably including mineralogy or petrology.

To be chosen for the College's Distinguished Majors program, students must achieve an overall GPA of 3.4 or above. This year, the department selected **Olivia Beidler**, **Kendall Colenbaugh**, **Brianna Handford**, **Caitlyn Kelley**, **Julianne Kirby**, **Rachel Lombardo**, **Caroline Parmentier**, **Virginia Ruhland-Mauhs**, and **Victoria Thompson** as distinguished majors.

Graduate Students

Kinsey Tedford was the winner of the Environmental Sciences Student Excellence Award, the department's premier award. Dr. F. Gordon Tice established the award in 1992 to foster environmental research and scholarship; it recognizes and honors outstanding undergraduate or graduate students for their contributions to environmental sciences, their ability to communicate their findings, and their efforts to promote a better understanding of the environment.

The department offers a series of awards honoring exceptional graduate students in individual environmental sciences. **Chi Hang "Henry" Yeung** earned the Graduate Award in Ecology, **Hanne Borstlap** the Graduate Award in Hydrology, **Sixuan Zhang** the Graduate Award in Atmospheric Sciences, and **Ariana Flournoy** the Arthur A. Pegau Award in Geoscience. **Paola Granados** received the Ellison-Edmonson Award in Interdisciplinary Studies.

Alexandrea Rice was this year's winner of the Joseph K. Roberts Award, given to a student who presents the most meritorious research paper at a national meeting.

Chi Hang "Henry" Yeung was one of 14 students from the College and Graduate School of Arts & Sciences selected to present their research at the 2024 Huskey Graduate Research Exhibition.

Santiago Munevar Garcia received the Jay Zieman Research Publication Award, named after the late Jay Zieman, former chair of the department.

Alexandrea Rice and **Mirella Shaban** won the Graduate Student Mentor Award.

The Exploratory Research Awards, based on merit, were initiated to help selected students conduct preliminary research leading to the development of a thesis or dissertation proposal. The recipients this year were **Paolo Granados**, **Robert Joseph**, **Stephanie Petrovick**, and **Preston Thompson**.

Staff

Laurie Hammond won the Department Chair's Award, which recognizes an individual who has performed extraordinary service to the department.

Faculty

Lawrence Band, the Ernest H. Ern Professor of Environmental Sciences, was coeditor of a special issue of the *Journal of Hydrology* titled "Non-Hortonian Processes in Urban Watershed." He is a member of the National Academies' committee reviewing approaches for managing pollutant loads in highway stormwater runoff and the proposal review committee for the National Science Foundation's Dynamics of Integrated Socio-Environmental Systems program. He also served as an external reviewer for promotion and tenure decisions at other universities. At UVA, he was a member of the Promotion and Tenure Committee of the School of Data Science.

Peter Berg reviewed National Science Foundation proposals and manuscripts for peer-reviewed journals. In addition, he arranged for a session on Underwater Flux Measurements in the Benthic Environment for the Association for the Sciences of Limnology and Oceanography Aquatic Sciences Meeting in Spain.

David Carr was an associate editor of the *American Journal of Botany* and an ad hoc reviewer for *New Phytologist*, *Ecological Restoration*, and *Annals of Botany*. For the UVA College and Graduate School of Arts & Sciences, he served as a member of its Chairs and Directors Committee as well as its General Faculty Promotion and Renewal Committee. This year, Carr received a Distinguished Scholars Award from the University of Virginia's Office of Engagement and Lifetime Learning.

Max Castorani served on the editorial board of *Ecosphere* and as an external advisor for the Plum Island Ecosystems Long Term Ecological Research project. He also served as an external thesis examiner and orals examiner for a doctoral student at the University of Auckland Institute of Marine Science.

Robert Davis, the director of undergraduate programs, reviewed manuscripts for *OENO One*, *International Journal of Biometeorology* and *Weather, Climate, and Society* and chaired a session titled "Human Health and Epidemiology" at the 23rd International Congress of Biometeorology. At the University, he chaired the Assembly Group of the Processions Committee and was an at-large faculty representative to the Raven Society and a reviewer for its Pete Cone Memorial Scholarship.

Stephan De Wekker was the editor of the *Journal of Applied Meteorology and Climatology* as well as an associate editor of *Atmosphere*. He was a member of the steering committee for the Transport and Exchange over Mountains–Programme and Experiment (TEAMx) as well as the steering committee for the Convective Organization and Venting Experiment in Complex Terrain (CONVECT). De Wekker reviewed grant proposals for the National Science Foundation and the German Science Foundation, wrote numerous journal reviews, and was an external reviewer for a promotion and tenure decision. At the University, he chaired the Faculty Senate Nominating Committee and represented UVA at the University Corporation for Atmospheric Research.

Scott Doney, the Joe D. and Helen J. Kington Professor in Environmental Change, is a Web of Science Clarivate Analytics Highly Cited Researcher in Cross-Field research and a Research.com Earth Science in the United States Leader Award winner. He served as assistant director for ocean climate science and policy in the White House Office of Science. In this position, he was the co-lead of the White House's Ocean Climate Action Plan and the Ocean Justice Strategy.

At the University, he was a storyteller in a program titled, "The Story Collider Presents: Stories from University of Virginia," jointly sponsored by the Office of the President, the Office of the Provost, and Story Collider, and was interviewed by the Office of the Vice President for Research for a research spotlight shared on social media.

Howard E. Epstein was the chair of the Department of Environmental Sciences and the Sidman P. Poole Professor of Environmental Sciences. He was a co-lead of the Arctic Observing Working Group, an initiative of the National Science Foundation's (NSF's) Navigating the New Arctic program, and an institutional representative to the Arctic Research Consortium. In addition, he was co-chair of the Vegetation Dynamics Working Group, part of the National Aeronautics and Space Administration's Arctic-Boreal Vulnerability Experiment, as well as of the Environmental Working Group of the Digital Belt and Road Initiative, sponsored by the Chinese Academy of Sciences. He co-organized the Symposium on Arctic Infrastructure at the Arctic Science Summit Week as well as the Permafrost & Infrastructure Symposium held in Utqiaġvik, Alaska. He was a conference award judge for the American Geophysical Union and reviewed numerous journal articles and an NSF proposal.

At the University, Epstein was codirector of the College Science Scholars program, a member of the EXPAND Fellowship Steering Committee, and a member of the Global Sustainability Advisory Committee, part of the Public Service Pathways program. This year, he was singled out for a Research Collaboration Award by UVA's Office of the Vice President for Research. Epstein won the department's Graduate Student Association Award, which recognizes members of the department who have been particularly helpful to the graduate student body.

Kevin Grise was a member of the United States Climate Variability and Predictability (U.S. CLIVAR) Program Process Study and Model Improvement Panel and was a member of the organizing committee for the U.S. CLIVAR workshop on Confronting Earth System Model Trends and Observations. In addition, he was selected as one of 18 new editors for the journal *Atmospheric Chemistry and Physics*. He served as a peer reviewer for numerous journal articles—he received the Outstanding Reviewer Award from *Geophysical Research Letters*—as well as for a funding proposal for the National Science Foundation. At the University, Grise was named an Advance Fellow in the College and Graduate School of Arts & Sciences faculty-led STEM Student Success Initiative. He represented UVA at the University Corporation for Atmospheric Research. This year, Grise won an Award for Excellence in Teaching from the Jefferson Scholars Foundation.


ALEXIS TILOT

Kyle Haynes was on the editorial boards of *Ecography* and *Oecologia* and was a guest editor for a special edition of *Current Opinion in Insect Science*. He also reviewed manuscripts for a number of professional journals and proposals for the National Science Foundation.

Manuel Lerdau was the editor-in-chief of *Ecosphere*, an associate editor of *Biology Letters and Ecology*, and guest subject editor of *Ecological Applications*. He served on the National Science Foundation Graduate Research Fellowship Program panel and reviewed proposals for its Division of Integrative Organismal Systems. In addition, he reviewed manuscripts for several professional journals and served on the publications committee of the Ecological Society of America. At the University, Lerdau chaired the College and Graduate School of Arts & Sciences Faculty Rules Committee. In addition, he was a member of the Morven Steering Committee, the Nelson Fund Committee, the Provost's Working Group on Faculty Evaluation, the advisory board of the Global Infectious Disease Institute, the Sustainability@ UVA Committee, the UVA Food Collaborative Steering Committee, and the Southeast Asia Studies Committee. He was also a faculty mentor for the Grad STAR Faculty Student Mentoring Program.

Ajay Limaye won a National Science Foundation (NSF) CAREER Award. He reviewed proposals for the NSF and the National Aeronautics and Space Administration and manuscripts for a variety of professional journals. This year, the department awarded him its Tice Prize for research excellence.

Stephen A. Macko was a visiting scholar at the Smithsonian Institution. He was editor-in-chief of *Nitrogen*, section editor-in-chief of *Geosciences* (*Biogeosciences*), and served on the editorial board of *Minerals*. He was a convener for the Geosciences Information for Teachers and the Science in Tomorrow's Classroom virtual workshops, held in conjunction with the European Geosciences Union (EGU) General Assembly. He was also a member of the EGU's Committee on Education. Macko was on the Major Instrumentation Panel at the National Science Foundation and served as an external reviewer for a promotion and tenure decision. He was included in *American Men and Women in Science*, *Who's Who in America, Science and Engineering*, *Who's Who in America, Science Education*, and *Who's Who in the World, Science and Engineering*. At the University, Macko was a member of the Faculty Advisory Committee to the Honor Committee, the Summer Session Advisory Committee, and the University Libraries Committee. He was a Hereford College Faculty Fellow.

Antonios Mamalakis is an associate editor of *Artificial Intelligence for the Earth Systems* and *Stochastic Environmental Research and Risk Assessment*. He reviewed numerous articles for professional journals.

Karen J. McGlathery, the Sherrell J. Aston Chaired Professor of Environmental Sciences, is the lead principal investigator of the Virginia Coast Reserve Long-Term Ecological Research (LTER) program and director of UVA's Environmental Institute. She sat on the LTER National Science Council and the Advisory Committee of the Florida Coastal Everglades LTER. In addition, she was a member of the National Science Foundation (NSF) Coastlines and People (CoPe) Award Steering Group, the NSF Task Force on Climate Equity, the Governor's Technical Advisory Committee for Virginia's Coastal Resilience Master Plan, and the Research and Education Advisory Council of Virginia Sea Grant. She was also an associate editor of *Ecosystems*.

McGlathery serves the University in a number of capacities. In addition to directing the Environmental Institute, she was codirector of University's Coastal Conservatory Environmental Humanities Consortium. She also served on the Global Sustainability Advisory Committee, part of the Public Service Pathways program, and the Extramural Funding and Research Administration Working Group, part of the Strategic Research Infrastructure Initiative under the auspices of the Office of the Vice President for Research. In addition, she was advisor to the provost and vice president of research on Environmental Resilience and Sustainability Grand Challenge Investments and was appointed by the provost to chair the Environmental Grand Challenges Faculty Hiring Coordinating Committee.

Lauren Miller was secretary of the American Geophysical Union Cryospheric Sciences Section and was appointed to the National Academies of Sciences, Engineering, and Medicine Polar Research Board. She served as review editor of the cryospheric sciences section of *Frontiers of Earth Science* and as an associate editor of a special issue of Earth Surface Processes and Landforms on "Glaciated Landscapes: Geomorphology as a Tool for Understanding Past, Present, and Future Glacier and Ice Sheet Behaviour." This year, she was a visiting fellow at Hokkaido University. Miller was also a member of the Louis Stokes Alliance for Minority Participation (LSAMP) Virginia-North Carolina Alliance Governing Board.

At the University, she co-chaired the Tribal Liaison Search Committee, was the Native and Indigenous relations community representative to the Chairs Summit, a member of the Native and Indigenous Relations Community Steering Committee, and a member of the Indigenous Studies Working Group of the Democracy Initiative. She was a college fellow for the University's Engagement Program, faculty advisor for UVA chapter of Epsilon Eta, UVA representative for environmental sciences curriculum development for Transfer Virginia, and an Interdisciplinary PhD in Indigenous Studies STEM mentor. This year, Miller won the Award for Excellence in Teaching from the Jefferson Scholars Foundation and was the recipient of an Outstanding Researcher Award from the Office of the Vice President for Research.

Charity Nyelele was a member of the Ecosystem Service Partnership and a fellow on the Intergovernmental Platform on Biodiversity and Ecosystem Services thematic assessment of the interlinkages among biodiversity, water, food, and health. At the University, she was recognized by first- through third-year undergraduate students as "the one individual who helped them the most with their career development" through a survey conducted by the UVA Career Center.

Michael Pace, the W. W. Corcoran Professor of Natural History and director of graduate studies for the department, served as the interim treasurer of the Association for the Sciences of Limnology and Oceanography and co-facilitator of a Career Planning Workshop for Early-Career Environmental Scientists at King Abdullah University of Science and Technology. He was also the William V. Kaeser Visiting Scholar at the University of Wisconsin. At the University, Pace was a member of the Endowed Professors Committee in the College and Graduate School of Arts & Sciences.

John Porter participated in several working groups of the Environmental Data Initiative, including the Units Working Group, the Ecological Metadata Language Best Practices Working Group, and the Faceted Search Working Group. He was also an active member in the Envirosensing Cluster of the Earth Science Information Partnership.

Sally Pusede was a chapter coauthor on air quality for the Fifth National Climate Assessment and a peer reviewer for numerous articles. At the University, she was a member of the Steering Committee for the Strategic Plan for Research, Scholarship, and Creative Activity.

Matthew Reidenbach was a working group member of the Advancing Public Engagement Across LTERs program. This year, Reidenbach was the recipient of an Outstanding Researcher Award from the Office of the Vice President for Research.

Justin Richardson served as a proposal reviewer for the U.S. National Science Foundation, the Swiss National Science Foundation, and the Canadian Foundation for Innovation and as an associate editor of *Biogeochemistry*. For the department, he was a member of the Graduate Academic Review Committee and the Goodell Fund Committee.

T'ai Roulston was a subject editor of *Ecosphere* and *Insects*.

Todd Scanlon is the associate chair of the department. He was a member of the award committee for the American Geophysical Union Hydrologic Sciences Early Career Award and a proposal reviewer for the National Science Foundation. At the University, he served as a member of the College and Graduate School of Arts & Sciences Committee on Faculty Rules and was a selection committee member of the Rising Scholars Postdoctoral Fellows program.

Kathleen Schiro was an associate editor of *Monthly Weather Review* and reviewed manuscripts for a number of professional journals.

Patricia Wiberg was a member of the Fellows Selection Committee of the American Geophysical Union's Earth and Planetary Surface Processes Focus Group. She served on the Steering Committee of the National Science Foundation-sponsored Community Sediment Dynamics Modeling System and on the advisory board of the Sediment Workgroup, part of the Regional Monitoring Program for Water Quality in San Francisco Bay.

Xi Yang served on the Steering Committee for Flux Course, a two-week educational program for graduate students sponsored by the AmeriFlux Network, and the Guidance Team of the Spectral Ecology Summer School to ensure an equitable and inclusive selection process. At the University, he served on the Environmental Institute Proposal Review Committee.

Published Peer-Reviewed Papers, Book Chapters, and Books by Faculty, Staff, and Graduate Students for Summer 2023–Spring 2024

Afshar-Mohajer, N., & **Shaban, M.** (2024). Source Tracing of PM2. 5 in a Metropolitan Area Using a Low-Cost Air Quality Monitoring Network: Case Study of Denver, Colorado, USA. *Atmosphere, 15*(7), 797. https://doi.org/10.3390/atmos15070797

Ahmed, F., Neelin, J. D., Hill, S. A., **Schiro, K. A.**, & Su, H. (2023). A process model for ITCZ narrowing under warming highlights clear-sky water vapor feedbacks and gross moist stability changes in AMIP models. *Journal of Climate, 36*(15), 4913–4931. https://doi.org/10.1175/JCLI-D-22-0689.1.

McGovern, A., Gagne, D. J., Wirz, C. D., Ebert-Uphoff, I., Bostrom, A., Rao, Y., **Mamalakis, A.**, ... & Peterson, T. (2023). Trustworthy Artificial Intelligence for Environmental Sciences: An Innovative Approach for Summer School. *Bulletin of the American Meteorological Society, 104*(6), E1222-E1231. https://doi.org/10.1175/BAMS-D-22-0225.1.

Bandou, D., Schlunegger, F., Kissling, E., Marti, U., Reber, R., & Pfander, J. (2023). Overdeepenings in the Swiss plateau: U-shaped geometries underlain by inner gorges. *Swiss journal of geosciences, 116*(1), 19. https://doi.org/10.1186/s00015-023-00447-y.

Bansal, S., Creed, I. F., Tangen, B. A., Bridgham, S. D., Desai, A. R., Krauss, K. W., **Berg, P.**, ... & Zhu, X. (2023). Practical guide to measuring wetland carbon pools and fluxes. *Wetlands, 43*(8), 105. https://doi: 10.1007/s13157-023-01722-2.

Berg, P., & Huettel, M. (2023). Coastal marine megaripple fields are metabolic hotspots with highly dynamic oxygen exchange. *Limnology and Oceanography Letters, 8*(6), 867–875. https://doi: 10.1002/lol2.10345.

Bourbonnais, A., Chang, B. X., Sonnerup, R. E., **Doney, S. C.**, & Altabet, M. A. (2023). Marine N_2O cycling from high spatial resolution concentration, stable isotopic and isotopomer measurements along a meridional transect in the eastern Pacific Ocean. *Frontiers in Marine Science, 10*, 1137064. https://doi.org/10.3389/fmars.2023.1137064.

Buck, A. J., Zarrella-Smith, K. A., Jordaan, A., & **Richardson, J. B.** (2024). Ecotoxicology of Trace Elements in European Oyster (Ostrea edulis), Seawater, and Sediments across Boston Harbor, Massachusetts, USA. *Bulletin of Environmental Contamination and Toxicology, 112*(1), 20. https://doi: 10.1007/s00128-023-03844-z.

Buelo, C. D., Besterman, A. F., Walter, J. A., **Pace, M. L.**, Ha, D. T., & **Tassone, S. J.** (2024). Quantifying disturbance and recovery in estuaries: Tropical cyclones and high-frequency measures of oxygen and salinity. *Estuaries and Coasts, 47*(1), 18–31. https://doi.org/10.1007/s12237-023-01255-1.

Cannon, C. H., & **Lerdau, M.** (2023). Conservation should not make 'perfect' an enemy of 'good'. *Trends in Plant Science, 28*(9), 971–972. https://doi.org/10.1016/j.tplants.2023.06.010.

Carvalho, L. M., Duine, G. J., Clements, C., **De Wekker, S. F.**, Fernando, H. J., Fitzjarrald, D. R., ... & Vömel, H. (2024). The Sundowner Winds Experiment (SWEX) in Santa Barbara, California: Advancing Understanding and Predictability of Downslope Windstorms in Coastal Environments. *Bulletin of the American Meteorological Society, 105*(3), E532–E558. https://doi.org/10.1175/BAMSD-22-0171.1.

Castiblanco, E. S., Groffman, P. M., Duncan, J., **Band, L. E.**, Doheny, E., Fisher, G. T., ... & Suchy, A. K. (2023). Long-term trends in nitrate and chloride in streams in an exurban watershed. *Urban Ecosystems, 26*(3), 831–844. https://doi.org/10.1007/s11252-023-01340-0.

Crump, B. C., **Blum, L. K.**, & **Mills, A. L.** (2023). Estuarine microbial ecology. In Day Jr., J. W., Kemp, W. M., Yáñez-Arancibia, A., & Crump, B. C. (Eds.). (2012). *Estuarine ecology*. John Wiley & Sons.

Davis, R. E., Roney, P. C., Pane, M. M., Johnson, M. C., Leigh, H. V., Basener, W., ... & Novicoff, W. M. (2023). Climate and human mortality in Virginia, 2005–2020. *Science of The Total Environment, 894*, 164825. https://doi.org/10.1016/j.scitotenv.2023.164825.

DeVries, T., Yamamoto, K., Wanninkhof, R., Gruber, N., Hauck, J., Müller, J. D., **Doney, S. C.**, ... & Zeng, J. (2023). Magnitude, trends, and variability of the global ocean carbon sink from 1985 to 2018. *Global Biogeochemical Cycles, 37*(10), e2023GB007780. https://doi.org/10.1029/2023GB007780.

Dressel, I. M., **Zhang, S.**, **Demetillo, M. A. G.**, Yu, S., Fields, K., Judd, L. M., ... & **Pusede, S. E.** (2024). Neighborhood-Level Nitrogen Dioxide Inequalities Contribute to Surface Ozone Variability in Houston, Texas. *ACS Es&t Air, 1*(9), 973–988. https://doi: 10.1021/acsestair.4c00009

Duine, G. J., **De Wekker, S. F.**, & Knievel, J. C. (2024). The Influence of Terrain Smoothing on Simulated Convective Boundary-Layer Depths in Mountainous Terrain. *Atmosphere, 15*(2), 145. https://doi.org/10.3390/atmos15020145.

Ewers Lewis, C. J., & **McGlathery, K. J.** (2023). A novel subsurface sediment plate method for quantifying sediment accumulation and erosion in seagrass meadows. *Frontiers in Marine Science, 10*, 1232619. https://doi.org/10.3389/fmars.2023.1232619.

Fauvel, C., Fuhrman, J., Ou, Y., Shobe, W., **Doney, S.**, McJeon, H., & Clarens, A. (2023). Regional implications of carbon dioxide removal in meeting net zero targets for the United States. *Environmental Research Letters, 18*(9), 094019. https://doi.org/10.1088/1748-9326/aced18.

Fritzeen, W. E., O'Rourke, P. R., Fuhrman, J. G., Colosi, L. M., Yu, S., Shobe, W. M., **Doney, S. C.**, ... & Clarens, A. F. (2023). Integrated Assessment of the Leading Paths to Mitigate CO_2 Emissions from the Organic Chemical and Plastics Industry. *Environmental Science & Technology, 57*(49), 20571–20582. https://doi.org/10.1021/acs.est.3c05202.

Frost, G. V., Macander, M. J., Bhatt, U. S., Berner, L. T., Bjerke, J. W., ... and **Epstein, H. E.** (2023). Tundra greenness. *Bulletin American Meteorological Society, 104*(9), S281–S284. https://edepot.wur.nl/583940.

Frost, G. V., Macander, M. J., Bhatt, U. S., Berner, L. T., Bjerke, J. W., **Epstein, H. E.**, ... & Yang, D. (2023). NOAA Arctic Report Card 2023: Tundra Greenness. https://doi:10.25923/s86a-jn24.

Frost, G. V., Bhatt, U. S., Macander, M. J., **Epstein, H. E.**, Raynolds, M. K., Waigl, C. F., & Walker, D. A. (2023). Eyes of the world on a warmer, less frozen, and greener Arctic. *Global Change Biology, 29*(16), 4453–4455. https://doi.org/10.1111/gcb.16767.

Fuhrman, J., Bergero, C., Weber, M., Monteith, S., Wang, F. M., Clarens, A. F., **Doney, S. C.**, ... & McJeon, H. (2023). Diverse carbon dioxide removal approaches could reduce impacts on the energy–water–land system. *Nature Climate Change, 13*(4), 341–350. https://doi.org/10.1038/s41558-023-01604-9.

Gohlke, J. M., Harris, M. H., Roy, A., Thompson, T. M., DePaola, M., Alvarez, R. A., **Pusede, S. E.**, ... & Vogel, S. A. (2023). State-of-the-science data and methods need to guide place-based efforts to reduce air pollution inequity. *Environmental Health Perspectives, 131*(12), 125003. https://doi:10.1289/EHP13063.

Granville, K. E., **Berg, P.**, & Huettel, M. (2023). A high-resolution submersible oxygen optode system for aquatic eddy covariance. *Limnology and Oceanography: Methods, 21*(3), 152–163. https://doi: 10.1002/lom3.10535.

Groffman, P. M., Suchy, A. K., Locke, D. H., Johnston, R. J., Newburn, D. A., **Band, L. E.**, ... & Zawojska, E. (2023). Hydro-bio-geo-socio-chemical interactions and the sustainability of residential landscapes. *PNAS nexus, 2*(10), pgad316. https://doi.org/10.1093/pnasnexus/pgad316.

Guan, Y., Keppel-Aleks, G., **Doney, S. C.**, Petri, C., Pollard, D., Wunch, D., ... & Té, Y. (2023). Characteristics of Interannual Variability in Space-based XCO 2 Global Observations. *EGUsphere, 2023*, 1–30. https://doi.org/10.5194/acp-23-5355-2023.

Guo, H., Goulden, M., Chung, M. G., **Nyelele, C.**, Egoh, B., Keske, C., ... & Bales, R. (2023). Valuing the benefits of forest restoration on enhancing hydropower and water supply in California's Sierra Nevada. *Science of The Total Environment, 876*, 162836. https://doi.org/10.1016/j.scitotenv.2023.162836

Lim, H. C., Lambrecht, D., Forkner, R. E., & **Roulston, T. H.** (2023). Minimal sharing of nosematid and trypanosomatid parasites between honey bees and other bees, but extensive sharing of Crithidia between bumble and mason bees. *Journal of Invertebrate Pathology, 198*, 107933. https://doi.org/10.1016/j.jip.2023.107933

Halpern, B. S., Boettiger, C., Dietze, M. C., Gephart, J. A., Gonzalez, P., Grimm, N. B., **Haynes, K. J.**, ... & Youngflesh, C. (2023). Priorities for synthesis research in ecology and environmental science. *Ecosphere, 14*(1), e4342. https://doi.org/10.1002/ecs2.4342.

Hanley, M. L., Vukicevich, E., **Rice, A. M.**, & **Richardson, J. B.** (2024). Uptake of toxic and nutrient elements by foraged edible and medicinal mushrooms (sporocarps) throughout Connecticut River Valley, New England, USA. *Environmental Science and Pollution Research, 31*(4), 5526-5539. https://doi.org/10.1007/s11356-023-31290-1.

Hardison, S. B., **McGlathery, K. J.**, & **Castorani, M. C.** (2023). Effects of seagrass restoration on coastal fish abundance and diversity. *Conservation Biology, 37*(6), e14147. https://doi.org/10.1111/cobi.14147 6.

Hauck, J., Gregor, L., Nissen, C., Patara, L., Hague, M., Mongwe, P., **Doney, S. C.**, ... & Terhaar, J. (2023). The Southern Ocean carbon cycle 1985–2018: Mean, seasonal cycle, trends, and storage. *Global Biogeochemical Cycles, 37*(11), e2023GB007848. https://doi.org/10.1029/2023GB007848.

Haynes, K. J., Liebhold, A. M., & Johnson, D. M. (2023). Editorial overview: Arthropod population dynamics at regional scales: Novel approaches and emerging insights. *Current Opinion in Insect Science, 57*, 101030. https://doi:10.1016/j.cois.2023.101030.

Haynes, K. J., Miller, G. D., Serrano-Perez, M. C., Hey, M. H., & Emer, L. K. (2023). Artificial light at night increases the nighttime feeding of monarch butterfly caterpillars without affecting host plant quality. *Basic and Applied Ecology, 72*, 10–15. https://doi.org/10.1016/j.baae.2023.07.007.

Heffernan, E., **Epstein, H. E.**, McQuinn, T. D., Rogers, B. M., Virkkala, A. M., Lutz, D., & **Armstrong, A.** (2024). Comparing assumptions and applications of dynamic vegetation models used in the Arctic-Boreal zone of Alaska and Canada. *Environmental Research Letters, 19*(9), 093003. https://doi.org/10.1088/1748-9326/ad6619.

Hendricks, A. S., Bhatt, U. S., Frost, G. V., Walker, D. A., Bieniek, P. A., Raynolds, M. K., **Epstein, H. E.**, ... & Comiso, J. C. (2023). Decadal Variability in Spring Sea Ice Concentration Linked to Summer Temperature and NDVI on the Yukon–Kuskokwim Delta. *Earth Interactions, 27*(1), e230002. https://doi.org/10.1175/EI-D-23-0002.1.

Herbert, L. C., **Lepp, A. P.**, Garcia, S. M., Browning, A., **Miller, L. E.**, Wellner, J., ... & Sherrell, R. M. (2023). Volcanogenic fluxes of iron from the seafloor in the Amundsen Sea, West Antarctica. *Marine Chemistry, 253*, 104250. https://doi.org/10.1016/j.marchem.2023.104250.

Herbst, R. S., Culver, T. B., **Band, L. E.**, Wilson, B., & Quinn, J. D. (2023). Integrating social equity into multiobjective optimization of urban stormwater low-impact development. *Journal of Water Resources Planning and Management, 149*(8), 04023038. https://doi.org/10.1061/JWRMD5.WRENG-5981.

Huang, S. Y., Lai, S. Y., **Limaye, A. B.**, Foreman, B. Z., & Paola, C. (2023). Confinement width controls the morphology and braiding intensity of submarine braided channels: Insights from physical experiments. *Earth Surface Dynamics Discussions, 2023*, 1–25. https://doi.org/10.5194/esurf-2023-6.

Huelsman, K., **Epstein, H.**, **Yang, X.**, Mullori, L., Červená, L., & Walker, R. (2023). Spectral variability in fine-scale drone-based imaging spectroscopy does not impede detection of target invasive plant species. *Frontiers in Remote Sensing, 3*, 1085808. https://doi: 10.3389/frsen.2022.1085808.

Hunt, L., Lhotáková, Z., Neuwirthová, E., Klem, K., Oravec, M., **Epstein, H.E.**, ... & Albrechtová, J. (2023). Leaf Functional Traits in Relation to Species Composition in an Arctic–Alpine Tundra Grassland. *Plants, 12*(5), 1001. https://doi:10.3390/plants12051001

Hutchins, M., Sweetman, A., Barry, C., **Berg, P.**, George, C., Pickard, A., & Qu, Y. (2023). MAKING WAVES: Effluent to estuary: Does sunshine or shade reduce downstream footprints of cities?. *Water Research, 247*, 120815. https://doi: 10.1016/j.watres.2023.120815.

Hwang, T., **Band, L. E.**, Oishi, A. C., & Kang, H. (2023). Greenup variability impact on seasonal streamflow and soil moisture dynamics in humid, temperate forests. *Water Resources Research, 59*(12), e2022WR034125. https://doi.org/10.1029/2022WR034125.

Jenkins, W. J., **Doney, S. C.**, Seltzer, A. M., German, C. R., Lott III, D. E., & Cahill, K. L. (2023). A North Pacific meridional section (US GEOTRACES GP15) of helium isotopes and noble gases I: Deep water distributions. *Global Biogeochemical Cycles, 37*(5), e2022GB007667. https://doi.org/10.1029/2022GB007667

Johnson, D. M., & **Haynes, K. J.** (2023). Spatiotemporal dynamics of forest insect populations under climate change. *Current Opinion in Insect Science, 56*, 101020. https://doi:10.1016/j.cois.2023.101020.

Kelleher, M. K., **Grise, K. M.**, & Schmidt, D. F. (2023). Variability in Projected North American Mean and Extreme Temperature and Precipitation Trends for the 21st Century: Model-To-Model Differences Versus Internal Variability. *Earth's Future, 11*(3), e2022EF003161. https://doi.org/10.1029/2022EF003161.

Khosronejad, A., **Limaye, A. B.**, Zhang, Z., Kang, S., Yang, X., & Sotiropoulos, F. (2023). On the morphodynamics of a wide class of large-scale meandering rivers: insights gained by coupling LES with sediment-dynamics. *Journal of Advances in Modeling Earth Systems, 15*(3), e2022MS003257. https://doi:10.1029/2022MS003257.

Kim, H. H., Laufkötter, C., Lovato, T., **Doney, S. C.**, & Ducklow, H. W. (2023). Projected 21st-century changes in marine heterotrophic bacteria under climate change. *Frontiers in Microbiology, 14*, 1049579. https://doi.org/10.3389/fmicb.2023.1049579.

Kim, K. Y., **Scanlon, T.**, Bakar, S., & Lakshmi, V. (2023). Decoupling of Ecological and Hydrological Drought Conditions in the Limpopo River Basin Inferred from Groundwater Storage and NDVI Anomalies. *Hydrology, 10*(8), 170. https://doi: 10.3390/hydrology10080170, 2023.

Kim, S. Y., Choi, Y. J., Son, S. W., **Grise, K. M.**, Staten, P. W., An, S. I., ... & Shin, J. (2023). Hemispherically asymmetric Hadley cell response to CO2 removal. *Science advances, 9*(30), eadg1801. https://doi.org/10.1126/sciadv.adg1801.

Kopacz, M., Breeze, V., Kondragunta, S., Frost, G., Anenberg, S., **Pusede, S.E.**, ... & Kalluri, S. (2023). Global Atmospheric Composition Needs from Future Ultraviolet–Visible–Near-Infrared (UV–Vis–NIR) NOAA Satellite Instruments. *Bulletin of the American Meteorological Society, 104*(3), E623-E630. https://doi:10.1175/BAMS-D-22-0266.1, 2023.

Kupková, L., Červená, L., Potůčková, M., Lysák, J., Roubalová, M., Hrázský, Z., **Epstein, H.E.**, ... & Müllerová, J. (2023). Towards reliable monitoring of grass species in nature conservation: Evaluation of the potential of UAV and PlanetScope multi-temporal data in the Central European tundra. *Remote Sensing of Environment, 294*, 113645. https://doi: 10.1016/j.rse.2023.113645.

Lang, S. E., Luis, K. M., **Doney, S. C.**, Cronin-Golomb, O., & **Castorani, M. C.** (2023). Modeling Coastal Water Clarity Using Landsat-8 and Sentinel-2. *Earth and Space Science, 10*(7), e2022EA002579. https://doi.org/10.1029/2022EA002579.

LeCroy, K. A., Krichilsky, E., Grab, H. L., **Roulston, T. H.**, & Danforth, B. N. (2023). Spillover of chalkbrood fungi to native solitary bee species from non-native congeners. *Journal of Applied Ecology, 60*(6), 1067–1076. https://doi.org: 10.1111/1365-2664.14399.

Lerdau, M. T. (2023). Extraterrestrial life: back story for the control experiment. *Nature, 623*(7989), 916. https://doi.org/10.1038/d41586-023-03742-8.

Lerdau, M. T., Monson, R. K., & Ehleringer, J. R. (2023). The carbon balance of plants: economics, optimization, and trait spectra in a historical perspective. *Oecologia, 203*(3), 297–310. https://doi.org/10.1007/s00442-023-05458-y.

Li, Q., Meidan, D., Hess, P., Añel, J. A., Cuevas, C. A., **Doney, S.**, ... & Saiz-Lopez, A. (2023). Global environmental implications of atmospheric methane removal through chlorine-mediated chemistry-climate interactions. *Nature communications, 14*(1), 4045. https://doi.org/10.1038/s41467-023-39794-7.

Limaye, A. B., Adler, J. B., Moodie, A. J., Whipple, K. X., & **Howard, A. D.** (2023). Effect of Standing Water on Formation of Fan-Shaped Sedimentary Deposits at Hypanis Valles, Mars. *Geophysical Research Letters, 50*(4), e2022GL102367. https://doi.org/10.1029/2022GL102367.

Liu, L., Smith, J. R., **Armstrong, A. H.**, Alcaraz-Segura, D., **Epstein, H. E.**, Echeverri, A., ... & Chaplin-Kramer, R. (2023). Influences of Satellite Sensor and Scale on Derivation of Ecosystem Functional Types and Diversity. *Remote Sensing, 15*(23), 5593. https://doi.org/10.3390/rs15235593.

Liu, S., Yan, Z., Wang, Z., Serbin, S., Visser, M., Zeng, Y., **Yang, X.**, ... & Wu, J. (2023). Mapping foliar photosynthetic capacity in sub-tropical and tropical forests with UAS-based imaging spectroscopy: Scaling from leaf to canopy. *Remote Sensing of Environment, 293*, 113612. https://doi.org/10.1016/j.rse.2023.113612.

Liu, X., & **Grise, K. M.** (2023). Implications of warm pool bias in CMIP6 models on the Northern Hemisphere wintertime subtropical jet and precipitation. *Geophysical Research Letters, 50*(15), e2023GL104896. https://doi.org/10.1029/2023GL104896.

Lovett, D. H., & **Carr, D. E.** (2024). The potential for elevated soil salinity to enhance the ecological trap effect of roadside pollinator habitat. *Journal of Insect Conservation, 28*(1), 103–111. https://doi.org/10.1007/s10841-023-00526-3.

Maciute, A., Holovachov, O., Glud, R. N., Broman, E., **Berg, P.**, Nascimento, F. J., & Bonaglia, S. (2023). Reconciling the importance of meiofauna respiration for oxygen demand in muddy coastal sediments. *Limnology and Oceanography, 68*(8), 1895–1905. https://doi: 10.1002/lno.12393.

Mamalakis, A., Barnes, E. A., & Ebert-Uphoff, I. (2023). Carefully choose the baseline: Lessons learned from applying XAI attribution methods for regression tasks in geoscience. *Artificial Intelligence for the Earth Systems, 2*(1), e220058. https://doi.org/10.1175/AIES-D-22-0058.1.

Mamalakis, A., Barnes, E. A., & Hurrell, J. W. (2023). Using explainable artificial intelligence to quantify "climate distinguishability" after stratospheric aerosol injection. *Geophysical Research Letters, 50*(20), e2023GL106137. https://doi.org/10.1029/2023GL106137.

Marino, R., Hayn, M., Howarth, R. W., Giblin, A. E., **McGlathery, K. J.**, & **Berg, P.** (2023). Nitrogen fixation associated with epiphytes on the seagrass Zostera marina in a temperate lagoon with moderate to high nitrogen loads. *Biogeochemistry, 166*(3), 211–226. https://doi.org/10.1007/s10533-023-01083-2.

Mason, C.A., **Rice, K.C.**, & **Mills, A.L.** (2023). Selected inputs for examining the complex relations between climate and streamflow in the Mid-Atlantic region of the United States, 1981–2020. U.S. Geological Survey data release. https://doi.org/10.5066/P9ZJ8OLR.

McGlynn, D. F., Frazier, G., Barry, L. E., **Lerdau, M. T.**, **Pusede, S. E.**, & Isaacman-VanWertz, G. (2023). Minor contributions of daytime monoterpenes are major contributors to atmospheric reactivity. *Biogeosciences, 20*(1), 45–55. https://doi:10.5194/bg-20-45-2023.

McKee, D. C., **Doney, S. C.**, Della Penna, A., Boss, E. S., Gaube, P., & Behrenfeld, M. J. (2023). Biophysical Dynamics at Ocean Fronts Revealed by Bio-Argo Floats. *Journal of Geophysical Research: Oceans, 128*(3), e2022JC019226. https://doi.org/10.1029/2022JC019226.

McKenzie, M. A., **Miller, L. E.**, Simkins, L. M., Slawson, J. S., MacKie, E. J., & Wang, S. (2022). Differential impact of isolated topographic bumps on glacial ice flow and subglacial processes. *EGUsphere, 2022*, 1–12. https://doi.org/10.5194/tc-17-2477-2023.

Miller, B., **Wolfe, W.**, Gentry, J. L., Grewal, M. G., Highley, C. B., De Vita, R., ... & Caliari, S. R. (2023). Supramolecular fibrous hydrogel augmentation of uterosacral ligament suspension for treatment of pelvic organ prolapse. *Advanced Healthcare Materials, 12*(22), 2300086. https://doi.org/10.1002/adhm.202300086.

Mitchell, K. A., **Doney, S. C**., & Keppel-Aleks, G. (2023). Characterizing average seasonal, synoptic, and finer variability in orbiting Carbon Observatory-2 XCO2 across North America and adjacent ocean basins. *Journal of Geophysical Research: Atmospheres, 128*(3), e2022JD036696. https://doi.org/10.1029/2022JD036696.

Munevar Garcia, S., **Miller, L. E.**, Falcini, F. A., & Stearns, L. A. (2023). Characterizing bed roughness on the Antarctic continental margin. *Journal of Glaciology*. https://doi.org/10.1017/jog.2023.88.

Neyra, J. S., & **Davis, R. E.** (2024). The association between climate and emergency department visits for renal and urinary disease in Charlottesville, Virginia. *Environmental Research, 240*, 117525. https://doi.org/10.1016/j.envres.2023.117525.

Nyelele, C., Keske, C., Chung, M. G., Guo, H., & Egoh, B. N. (2023). Using social media data and machine learning to map recreational ecosystem services. *Ecological Indicators, 154*, 110606. https://doi.org/:10.1016/j.ecolind.2023.110606.

Ochoa-Hueso, R., Delgado-Baquerizo, M., Risch, A. C., Ashton, L., Augustine, D., **Epstein, H. E.**, ... & Bremer, E. (2023). Bioavailability of macro and micronutrients across global topsoils: Main drivers and global change impacts. *Global Biogeochemical Cycles, 37*(6), e2022GB007680. https://doi.org/10.1029/2022GB007680.

Oxley, C. C., & Jurgens, L. J. (2024). Material type affects the community composition and abundance of hard-substrate assemblages in a sedimentary Atlantic estuary. *Marine Ecology Progress Series, 727*, 35–47. https://doi.org/10.3354/meps14497.

Le, P. V., Randerson, J. T., Willett, R., Wright, S., Smyth, P., Guilloteau, C., **Mamalakis, A.**, ... & Foufoula-Georgiou, E. (2023). Climate-driven changes in the predictability of seasonal precipitation. *Nature communications, 14*(1), 3822. https://doi.org/10.1038/s41467-023-39463-9.

Pace, M. L., Carpenter, S. R., & Likens, G. E. (2024). Jonathan J. Cole (1953–2023). *Limnology and Oceanography Bulletin, 33*(1), 28–29. https://doi.org/10.1002/lob.10611.

Pasquarella, V. J., Morin, R. S., Liebhold, A. M., Siegert, N. W., Rodenberg, C. A., & **Haynes, K. J.** (2023). Detecting changes in forest condition at landscape scales using a Landsat-based harmonic modeling approach. 149–162. https://doi.org/10.2737/SRS-GTR-273-Chap9.

Pierrat, Z., Magney, T., **Yang, X**, Khan, A., & Albert, L. (2023). Ecosystem observations from every angle. EOS. https://doi.org/10.1029/2023EO230483.

Reuman, D. C., **Castorani, M. C.**, Cavanaugh, K. C., Sheppard, L. W., Walter, J. A., & Bell, T. W. (2023). How environmental drivers of spatial synchrony interact. *Ecography, 2023*(10), e06795. https://doi.org/10.1111/ecog.06795.

Rice, K. C., Mason, C. A., & **Mills, A. L.** (2023). Examining the complex relations between climate and streamflow in the mid-atlantic region of the United States. *Environmental Research Communications, 5*(12), 125008. https://doi.org/10.1088/2515-7620/ad13a4.

Roebber, P. J., **Grise, K. M.**, & Gyakum, J. R. (2023). The Histories of well-documented maritime cyclones as portrayed by an automated tracking method. *Monthly Weather Review, 151*(11), 2905–2924. https://doi.org/10.1175/MWR-D-22-0287.1.

Saby, L., Herbst, R. S., Goodall, J. L., Nelson, J. D., Culver, T. B., Stephens, E., ... & **Band, L. E.** (2023). Assessing and improving the outcomes of nonpoint source water quality trading policies in urban areas: A case study in Virginia. *Journal of Environmental Management, 345*, 118724. https://doi.org/10.1016/j.jenvman.2023.118724.

Saenz, B. T., **McKee, D. C.**, **Doney, S. C.**, Martinson, D. G., & Stammerjohn, S. E. (2023). Influence of seasonally varying sea-ice concentration and subsurface ocean heat on sea-ice thickness and sea-ice seasonality for a 'warm-shelf' region in Antarctica. *Journal of Glaciology, 69*(277), 1466–1482. https://doi:10.1017/jog.2023.36.

Schiro, K. A., & Su, H. (2023). Interactions Between the Tropical Atmospheric Overturning Circulation and Clouds in Present and Future Climates. *Clouds and Their Climatic Impacts: Radiation, Circulation, and Precipitation*, 223–238.

Schmidt, D. F., **Grise, K. M.**, & **Pace, M. L.** (2023). Does the 11-year solar cycle affect lake and river ice phenology?. *Plos one, 18*(12), e0294995. https://doi.org/10.1371/journal.pone.0294995.

Seipp, K. Q., Maurer, T., Elias, M., Saksa, P., Keske, C., **Nyelele, C.**, ... & Bales, R. (2023). A multi-benefit framework for funding forest management in fire-driven ecosystems across the Western US. *Journal of Environmental Management, 344*, 118270. https://doi.org/10.1016/j.jenvman.2023.118270.

Shao, Z., Xu, Y., Wang, H., Luo, W., Wang, L., Huang, Y., **Doney, S. C.**, ... & Luo, Y. W. (2023). Global oceanic diazotroph database version 2 and elevated estimate of global oceanic N_2 fixation. *Earth System Science Data, 15*(8), 3673–3709. https://doi.org/10.5194/essd-15-3673-2023.

Simkins, L. M., Greenwood, S. L., Winsborrow, M. C., Bjarnadóttir, L. R., & Lepp, A. P. (2022). Advances in understanding subglacial meltwater drainage from past ice sheets. *Annals of Glaciology, 63*(87–89), 83–87. https://doi.org/10.1017/aog.2023.16.

Smith, A. J., **McGlathery, K.**, **Pace, M. L.**, Ewers Lewis, C. J., **Doney, S. C.**, **Berg, P.**, ... & Kirwan, M. L. (2024). Compensatory mechanisms absorb regional carbon losses within a rapidly shifting coastal mosaic. *Ecosystems, 27*(1), 122–136. https://doi.org/10.1007/s10021-023-00877-7.

Smith, R. S., & **Castorani, M. C.** (2023). Meta-analysis reveals drivers of restoration success for oysters and reef community. *Ecological Applications, 33*(5), e2865. https://doi.org/10.1002/eap.2865.

Stephens, C. M., **Band, L. E.**, Johnson, F. M., Marshall, L. A., Medlyn, B. E., De Kauwe, M. G., & Ukkola, A. M. (2023). Changes in blue/green water partitioning under severe drought. *Water Resources Research, 59*(11), e2022WR033449. https://doi.org/10.1029/2022WR033449.

Suchy, A. K., Groffman, P. M., **Band, L. E.**, Duncan, J. M., Gold, A. J., Grove, J. M., ... & Zhang, R. (2023). Spatial and temporal patterns of nitrogen mobilization in residential lawns. *Ecosystems, 26*(7), 1524–1542. https://doi.org/10.1007/s10021-023-00848-y 3.

Tassone, M. S., **Epstein, H. E.**, **Armstrong, A. H.**, Bhatt, U. S., Frost, G. V., Heim, B., ... & Walker, D. A. (2024). Drivers of heterogeneity in tundra vegetation productivity on the Yamal Peninsula, Siberia, Russia. *Environmental Research: Ecology, 3*(1), 015003. https://doi.org/10.1088/2752-664X/ad220f.

Tassone, S. J., & **Pace, M. L.** (2024). Increased Frequency of Sediment Heatwaves in a Virginia Seagrass Meadow. *Estuaries and Coasts, 47*(3), 656–669. https://doi.org/10.1007/s12237-023-01314-7.

Walter, J. A., Emery, K. A., Dugan, J. E., Hubbard, D. M., **Bell, T. W.**, Sheppard, L. W., ... & **Castorani, M. C.** (2024). Spatial synchrony cascades across ecosystem boundaries and up food webs via resource subsidies. *Proceedings of the National Academy of Sciences, 121*(2), e2310052120. https://doi.org/10.1073/pnas.2310052120.

Walter, J. A., Coombs, N. J., & **Pace, M. L.** (2023). Synchronous variation of dissolved organic carbon in Adirondack lakes at multiple timescales. *Limnology and Oceanography Letters, 8*(4), 649-656. https://dx.doi.org/10.1002/lol2.10328.

Wang, Y., **Limaye, A. B.**, & Chadwick, A. J. (2024). Topography-based particle image velocimetry of braided channel initiation. *Water Resources Research, 60*(4), e2023WR035229. https://doi.org/10.1029/2023WR035229.

West, J. J., Nolte, C. G., Bell, M. L., **Pusede, S. E.**, Shindell, D. T., & Wilson, S. M. 2023. Chapter 14. Air Quality. In: *Fifth National Climate Assessment*. Crimmins, A.R., C.W. Avery, D.R. Easterling, K.E. Kunkel, B.C. Stewart, and T.K. Maycock, Eds. U.S. Global Change Research Program, Washington, DC, USA.

Wiberg, P. L., Temperature amplification and marine heatwave alteration in shallow coastal bays. *Frontiers in Marine Science*. 10:1129295. https://doi:10.3389/fmars.2023.1129295.

Williams, C.A., Alin, S., Butman, D. S. **Doney, S. C.**, et. al. 2023. Process and attribution studies to uncover mechanistic responses of the carbon cycle to natural and anthropogenic influences, Chapter 3.3 in *2022 North American Carbon Program Science Implementation Plan*, 71–96, https://www.nacarbon.org/nacp/implementation_plan.html.

Wisnoski, N. I., Andrade, R., **Castorani, M. C.**, Catano, C. P., Compagnoni, A., Lamy, T., ... & Sokol, E. R. (2023). Diversity–stability relationships across organism groups and ecosystem types become decoupled across spatial scales. *Ecology, 104*(9), e4136. https://doi.org/10.1002/ecy.4136 4.

Wolfe, W. H., Martz, T. R., Dickson, A. G., Goericke, R., & Ohman, M. D. (2023). A 37-year record of ocean acidification in the Southern California current. *Communications Earth & Environment, 4*(1), 406. https://doi.org/10.1038/s43247-023-01065-0.

Wu, G., Guan, K., Jiang, C., Kimm, H., Miao, G., **Yang, X.**, ... & Moore, C. E. (2023). Can upscaling ground nadir SIF to eddy covariance footprint improve the relationship between SIF and GPP in croplands? *Agricultural and Forest Meteorology, 338*, 109532. https://doi.org/10.1016/j.agrformet.2023.109532.

Yang, X., **Li, R.**, **Jablonski, A.**, Stovall, A., Kim, J., Yi, K., ... & **Lerdau, M.** (2023). Leaf angle as a leaf and canopy trait: Rejuvenating its role in ecology with new technology. *Ecology Letters, 26*(6), 1005–1020. https://doi.org/10.1111/ele.14215.

Zhang, R., **Band, L. E.**, & Groffman, P. M. (2023). Balancing upland green infrastructure and stream restoration to recover urban stormwater and nitrate load retention. *Journal of Hydrology, 626*, 130364. https://doi.org/10.1016/j.jhydrol.2023.130364.

ENVIRONMENTAL SCIENCES AT THE UNIVERSITY OF VIRGINIA

OUR MISSION
Our mission is to advance understanding of the environment through interdisciplinary scientific research, education, and service.

OUR VISION
We work at the forefront of research to discover, integrate, and communicate new knowledge and generate wisdom about environmental systems. We provide outstanding and innovative education and exceptional mentorship. We pursue fundamental, transformative science that provides objective and impactful information for management, policymaking, and environmental justice.

OUR VALUES
Our mission and vision are supported by a commitment to a core set of values that guide what we do as a community of scholars:

- **Integrity:** Maintain the highest standards of scientific, academic, and professional ethics
- **Diversity and Inclusivity:** Strengthen and foster a community that supports people from diverse backgrounds and empowers individuals for who they are
- **Freedom of Inquiry:** Promote open exploration of science that is accepting of respectful, constructive criticism and unbound by external pressures
- **Societal Impact:** Conduct science that makes a positive impact on the world
- **Collegiality:** Facilitate collaboration, teamwork, and support for each other's success
- **Community Engagement:** Respectfully and cooperatively engage with the communities where we live, study, and work

UNIVERSITY OF VIRGINIA
ENVIRONMENTAL SCIENCES
291 McCormick Road
P.O. Box 400123
Charlottesville, VA 22904-4123